

Cole Plagens

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EDUCATION

University of Houston

Houston, TX

Bachelor of Science in Computer Science, Minor in Mathematics, GPA: 3.64

Dec. 2025

- Relevant Coursework: Database Systems, Data Structures & Algorithms, Data Science I, Software Design
- Dean's List Fall 2023 - Fall 2025

PROJECTS

Incident Monitoring Platform

Feb. 2026 – Mar. 2026

- Designed and deployed a scalable log ingestion API using ASP.NET Core on AWS EC2, handling 10k+ simulated log events with robust validation, RESTful design, and unit/integration testing
- Implemented a fault-tolerant, asynchronous processing pipeline using RabbitMQ and Redis, reducing log processing latency by 40% and enabling real-time anomaly detection under load
- Engineered structured log storage in PostgreSQL (AWS RDS) and implemented statistical anomaly detection to identify error spikes and automatically generate incident alerts

Volunteer Activity Application

Sept. 2025 – Nov. 2025

- Led a team of four developers to build a full-stack volunteer management platform with a React frontend and PostgreSQL-backed REST API
- Coordinated task distribution, code reviews, and feature integration to deliver secure authentication and role-based access control
- Designed an automated volunteer-event matching system based on user profiles, improving coordination and user engagement through real-time notifications

Theme Park Management System

Jan. 2025 – Apr. 2025

- Led a team of five developers in building a full-stack management system using React and ASP.NET to support ticketing, ride browsing, and user accounts with role-based access
- Designed a relational MySQL database with constraints and optimized queries to maintain data integrity across system operations
- Developed admin dashboards with analytics for revenue and operations, integrating backend services and deploying via Azure and Vercel

EXPERIENCE

Crew Member

July 2022 – Aug. 2023; May 2025 – Present

Firehouse Subs

Houston, TX

- Collaborated in a fast-paced team environment to deliver efficient service, demonstrating reliability, communication, and time management under high demand

ACTIVITIES

CougarCS Tutor

Sept. 2025 – Dec. 2025

University of Houston

Houston, TX

- Debugged complex C/C++ implementations and guided students through algorithmic problem-solving
- Analyzed student codebases to identify logic errors, runtime issues, and inefficiencies, improving solution correctness and reinforcing best practices in debugging and optimization

TECHNICAL SKILLS

Languages: Python, C/C++, C#, TypeScript, JavaScript, SQL

Frameworks & Tools: React, ASP.NET Core, Node.js, Express, Tailwind CSS

Cloud & DevOps: AWS (EC2, RDS), Azure, Docker, Git, GitHub Actions, Jenkins (CI/CD Pipelines)

Databases: PostgreSQL, MySQL, Redis